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VAPOR GENERATION SYSTEM AND APPARATUS

Abstract

A vapor generation apparatus includes a fluid channel extending between a first end and a second end of the vapor generation apparatus, a plurality of first fluid passageways extending between the first end and the second end, and a plurality of second fluid passageways extending between the first end and the second end. The plurality of first fluid passageways are between the fluid channel and the plurality of second fluid passageways. The vapor generation apparatus also includes a fluid chamber adjacent the second end and configured to receive a first fluid and a separator adjacent the first end and in fluid communication with the fluid channel, the plurality of first fluid passageways, and the plurality of second fluid passageways. The fluid chamber is in fluid communication with the fluid channel and the plurality of first fluid passageways.

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Background/Summary

FIELD
[0001] The present disclosure relates to turbine engines including steam systems.
BACKGROUND
[0002] Turbine engines generally include a fan and a core section arranged in flow communication with one another. A combustor is arranged in the core section to generate combustion gases for driving a turbine in the core section of the turbine engine.
[0003] Turbine engines may also include a steam system to extract steam from the combustion gases. The extracted steam may drive a steam turbine connected to the turbine of the turbine engine. However, steam systems that improve the separation of the steam from liquid flowing though the steam system are desirable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:
[0005] FIG. **1** is a schematic cross-sectional view of a turbine engine according to an exemplary embodiment of the present disclosure.
[0006] FIG. **2** is a schematic diagram of the turbine engine of FIG. **1** and a steam system with a thermal transport system according to an exemplary embodiment of the present disclosure.
[0007] FIG. **3** is a schematic diagram of a vapor generation system of the turbine engine of FIGS. **1-2** according to an exemplary embodiment of the present disclosure.
[0008] FIG. **4**A is a perspective, detailed view of a separator of the vapor generation system of FIG. **3** according to an exemplary embodiment of the present disclosure.
[0009] FIG. **4**B is a cross-sectional view of the separator of the vapor generation system of FIG. **4**A according to an exemplary embodiment of the present disclosure.
[0010] FIG. **4**C is a detailed, cross-sectional view of the separator of the vapor generation system of FIG. **4**B according to an exemplary embodiment of the present disclosure.
[0011] FIG. **5**A is a perspective view of one of a plurality of separation passageways of the separator of FIGS. **4**A-**4**C according to an exemplary embodiment of the present disclosure.
[0012] FIG. **5**B is a rear, perspective view of the plurality of separation passageways of FIG. **5**A according to an exemplary embodiment of the present disclosure.
[0013] FIG. **6**A is a perspective view of the vapor generation system of FIG. **3** according to an exemplary embodiment of the present disclosure.
[0014] FIG. **6**B is a cross-sectional view of the vapor generation system of FIG. **6**A according to an exemplary embodiment of the present disclosure.
[0015] FIG. **7**A is a perspective view of a vapor generation system according to an exemplary embodiment of the present disclosure.
[0016] FIG. **7**B is a cross-sectional view of the vapor generation system of FIG. **6**A according to an exemplary embodiment of the present disclosure.
[0017] FIG. **8**A is a perspective view of another vapor generation system of the turbine engine of FIGS. **1-2** according to an exemplary embodiment of the present disclosure.
[0018] FIG. **8**B is a cross-sectional view of the vapor generation system of FIG. **8**A according to an exemplary embodiment of the present disclosure.
[0019] FIG. **8**C is a cross-sectional view of the vapor generation system of FIG. **8**A according to an exemplary embodiment of the present disclosure.
DETAILED DESCRIPTION
[0020] Reference will now be made in detail to present embodiments of the disclosure, one or more

examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

[0021] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

[0022] The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0023] The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.

[0024] The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.

[0025] The term “turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.

[0026] The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.

[0027] The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

[0028] As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the gas turbine engine. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the centerline of the gas turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the gas turbine engine.

[0029] As used herein, a “bypass ratio” of a turbine engine is a ratio of bypass air through a bypass of the turbine engine to core air through a core inlet of a turbomachine of the turbine engine. For example, the bypass ratio is a ratio of bypass air **62** entering the bypass airflow passage **56** to core air **64** entering the turbomachine **16**.

[0030] The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

[0031] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

[0032] For purposes of the description hereinafter, the terms “upper,” “lower,” “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” “lateral,” “longitudinal,” and derivatives thereof shall relate to the embodiments as they are oriented in the drawing figures. However, it is to be understood that the embodiments may assume various alternative variations, except where expressly specified to the contrary. It is also to be understood that the specific devices illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the disclosure. Hence, specific dimensions and other physical characteristics related to the embodiments disclosed herein are not to be considered as limiting.

[0033] As used herein, the term “complementary” with reference to a radius of curvature of two radii of curvature, refers to the two radii of curvature being equal, or a larger of two the radii of curvature being no more than 10% greater than a smaller of the two radii of curvature.

[0034] The term “adjacent” as used herein with reference to two walls and/or surfaces refers to the two walls and/or surfaces contacting one another, or the two walls and/or surfaces being separated only by one or more nonstructural layers and the two walls and/or surfaces and the one or more nonstructural layers being in a serial contact relationship (e.g., a first wall/surface contacting the one or more nonstructural layers, and the one or more nonstructural layers contacting a second wall/surface).

[0035] The present disclosure is generally related to turbine engines including steam systems. Within the steam system, steam is separated from liquid. In some example embodiments, the steam system includes a heat exchanger for separating the steam from the liquid. For example, the heat exchanger may include a separator, such as an inertial separator, for separating the liquid from the steam.

[0036] Referring now to the drawings, FIG. **1** is a schematic cross-sectional view of a turbine engine **10** according to an exemplary embodiment of the present disclosure.

[0037] In at least one example embodiment, the turbine engine **10** includes a steam system **100**. As shown in FIG. **1**, the turbine engine **10** has an axial direction A (extending parallel to a longitudinal centerline axis **12**) and a radial direction R that is normal to the axial direction A. In general, the turbine engine **10** includes a fan section **14** and a turbomachine **16** disposed downstream from the fan section **14**.

[0038] The turbomachine **16** includes an outer casing **18** that is substantially tubular and defines an annular core inlet **20**. As schematically shown in FIG. **1**, the outer casing **18** encases, in serial flow relationship, a compressor section **21** including a booster or a low-pressure compressor (“LPC”) **22** followed downstream by a high-pressure compressor (“HPC”) **24**, a combustor **26**, a turbine section **27**, including a high-pressure turbine (“HPT”) **28**, followed downstream by a low-pressure turbine (“LPT”) **30**, and one or more core exhaust nozzles **32**. A high-pressure (“HP”) shaft **34** or a spool drivingly connects the HPT **28** to the HPC **24** to rotate the HPT **28** and the HPC **24** in unison. The HPT **28** is drivingly coupled to the HP shaft **34** to rotate the HP shaft **34** when the HPT **28** rotates. A low-pressure (“LP”) shaft **36** drivingly connects the LPT **30** to the LPC **22** to rotate the LPT **30** and the LPC **22** in unison. The LPT **30** is drivingly coupled to the LP shaft **36** to rotate the LP shaft **36** when the LPT **30** rotates. The compressor section **21**, the combustor **26**, the turbine section **27**, and the one or more core exhaust nozzles **32** together define a working gas flow path **33**.

[0039] For the embodiment depicted in FIG. **1**, the fan section **14** includes a fan **38** (e.g., a variable pitch fan) having a plurality of fan blades **40** coupled to a disk **42** in a spaced apart manner. As depicted in FIG. **1**, the fan blades **40** extend outwardly from the disk **42** generally along the radial direction R. Each fan blade **40** is rotatable relative to the disk **42** about a pitch axis P by virtue of the fan blades **40** being operatively coupled to an actuator **44** configured to collectively vary the pitch of the fan blades **40** in unison. The fan blades **40**, the disk **42**, and the actuator **44** are together rotatable about the longitudinal centerline axis **12** via a fan shaft **45** that is powered by the LP shaft **36** across a power gearbox, also referred to as a gearbox assembly **46**. The gearbox assembly **46** is shown schematically in FIG. **1**. The gearbox assembly **46** includes a plurality of gears for adjusting the rotational speed of the fan shaft **45** and, thus, the fan **38** relative to the LP shaft **36**.

[0040] Referring still to the exemplary embodiment of FIG. **1**, the disk **42** is covered by a rotatable fan hub **48** aerodynamically contoured to promote an airflow through the plurality of fan blades **40**. In addition, the fan section **14** includes an annular fan casing or a nacelle **50** that circumferentially surrounds the fan **38** and/or at least a portion of the turbomachine **16**. The nacelle **50** is supported relative to the turbomachine **16** by a plurality of circumferentially spaced outlet guide vanes **52**. Moreover, a downstream section **54** of the nacelle **50** extends over an outer portion of the turbomachine **16** to define a bypass airflow passage **56** therebetween. The one or more core exhaust nozzles **32** may extend through the nacelle **50** and be formed therein. In this exemplary embodiment, the one or more core exhaust nozzles **32** include one or more discrete nozzles that are spaced circumferentially about the nacelle **50**. Other arrangements of the core exhaust nozzles **32** may be used including, for example, a single core exhaust nozzle that is annular, or partially annular, about the nacelle **50**.

[0041] During operation of the turbine engine **10**, a volume of air **58** enters the turbine engine **10** through an inlet **60** of the nacelle **50** and/or the fan section **14**. As the volume of air **58** passes across the fan blades **40**, a first portion of air (bypass air **62**) is directed or routed into the bypass airflow passage **56**, and a second portion of air (core air **64**) is directed or is routed into the upstream section of the working gas flow path **33**, or, more specifically, into the annular core inlet **20**. The ratio between the first portion of air (bypass air **62**) and the second portion of air (core air **64**) is known as a bypass ratio. In some embodiments, the bypass ratio is greater than 18:1, enabled by a steam system **100**, detailed further below. The pressure of the core air **64** is increased by the LPC **22**, generating compressed air **65**, and the compressed air **65** is routed through the HPC **24** and further compressed before being directed into the combustor **26**, where the compressed air **65** is mixed with fuel **67** and burned to generate combustion gases **66** (combustion products). One or more stages may be used in each of the LPC **22** and the HPC **24**, with each subsequent stage further compressing the compressed air **65**. The HPC **24** has a compression ratio greater than 20:1, preferably, in a range of 20:1 to 40:1. The compression ratio is a ratio of a pressure of a last stage of the HPC **24** to a pressure of a first stage of the HPC **24**. The compression ratio greater than 20:1 is enabled by the steam system **100**, as detailed further below.

[0042] The combustion gases **66** are routed into the HPT **28** and expanded through the HPT **28** where a portion of thermal energy and/or kinetic energy from the combustion gases **66** is extracted via sequential stages of HPT stator vanes **68** that are coupled to the outer casing **18** and HPT rotor blades **70** that are coupled to the HP shaft **34**, thus, causing the HP shaft **34** to rotate, thereby supporting operation of the HPC **24**. The combustion gases **66** are routed into the LPT **30** and expanded through the LPT **30**. Here, a second portion of thermal energy and/or the kinetic energy is extracted from the combustion gases **66** via sequential stages of LPT stator **72** that are coupled to the outer casing **18** and LPT rotor blades **74** that are coupled to the LP shaft **36**, thus, causing the LP shaft **36** to rotate, thereby supporting operation of the LPC **22** and rotation of the fan **38** via the gearbox assembly **46**. One or more stages may be used in each of the HPT **28** and the LPT **30**. The HPC **24** having a compression ratio in a range of 20:1 to 40:1 enables the HPT **28** to have a pressure expansion ratio in a range of 1.5:1 to 4:1 and the LPT **30** having a pressure expansion ratio in a range of 4.5:1 to 28:1.

[0043] The combustion gases **66** are subsequently routed through the one or more core exhaust nozzles **32** of the turbomachine **16** to provide propulsive thrust. Simultaneously with the flow of the core air **64** through the working gas flow path **33**, the bypass air **62** is routed through the bypass airflow passage **56** before being exhausted from a fan bypass nozzle **76** of the turbine engine **10**, also providing propulsive thrust. The HPT **28**, the LPT **30**, and the one or more core exhaust nozzles **32** at least partially define a hot gas path **78** for routing the combustion gases **66** through the turbomachine **16**.

[0044] As noted above, the compressed air **65** (the core air **64**) is mixed with the fuel **67** in the combustor **26** to generate a fuel and air mixture, and combusted, generating combustion gases **66** (combustion products). The fuel **67** can include any type of fuel used for turbine engines, such as, for example, sustainable aviation fuels (SAF) including biofuels, JetA, or other hydrocarbon fuels. The fuel **67** also may be a hydrogen-based fuel (H.sub.2), and, while hydrogen-based fuel may include blends with hydrocarbon fuels, the fuel **67** used herein is preferably unblended, and referred to herein as hydrogen fuel. In some embodiments, the hydrogen fuel may comprise substantially pure hydrogen molecules (i.e., diatomic hydrogen). The fuel **67** may also be a cryogenic fuel. For example, when the hydrogen fuel is used, the hydrogen fuel may be stored in a liquid phase at cryogenic temperatures.

[0045] The turbine engine **10** includes a fuel system **80** for providing the fuel **67** to the combustor **26**. The fuel system **80** includes a fuel tank **82** for storing the fuel **67** therein, and a fuel delivery assembly **84**. The fuel tank **82** can be located on an aircraft (not shown) to which the turbine engine **10** is attached. While a single fuel tank **82** is shown in FIG. **1**, the fuel system **80** can include any number of fuel tanks **82**, as desired. The fuel delivery assembly **84** delivers the fuel **67** from the fuel tank **82** to the combustor **26**. The fuel delivery assembly **84** includes one or more lines, conduits, pipes, tubes, etc., configured to carry the fuel **67** from the fuel tank **82** to the combustor **26**. The fuel delivery assembly **84** also includes a pump **86** to induce the flow of the fuel **67** through the fuel delivery assembly **84** to the combustor **26**. Steam injected directly into or upstream of the combustor **26** adds mass flow to the core air **64** such that less core air **64** is needed to produce the same amount of work through the turbine section **27**. In this way, the pump **86** pumps the fuel **67** from the fuel tank **82**, through the fuel delivery assembly **84**, and into the combustor **26**. The fuel system **80** and, more specifically, the fuel tank **82** and the fuel delivery assembly **84**, either collectively or individually, may be a fuel source for the combustor **26**.

[0046] In some embodiments, for example, when the fuel **67** is a hydrogen fuel, the fuel system **80** includes one or more vaporizers **88** (illustrated by dashed lines) and a metering valve **90** (illustrated by dashed lines) in fluid communication with the fuel delivery assembly **84**. In this example, the hydrogen fuel is stored in the fuel tank **82** as liquid hydrogen fuel. The one or more vaporizers **88** heat the liquid hydrogen fuel flowing through the fuel delivery assembly **84**. The one or more vaporizers **88** are positioned in the flow path of the fuel **67** between the fuel tank **82** and the combustor **26** and are located downstream of the pump **86**. The one or more vaporizers **88** are in thermal communication with at least one heat source, such as, for example, waste heat from the turbine engine **10** and/or from one or more systems of the aircraft (not shown). The one or more vaporizers **88** heat the liquid hydrogen fuel and convert the liquid hydrogen fuel into a gaseous hydrogen fuel within the one or more vaporizers **88**. The fuel delivery assembly **84** directs the gaseous hydrogen fuel into the combustor **26**.

[0047] The metering valve **90** is positioned downstream of the one or move vaporizers **88** and the pump **86**. The metering valve **90** receives hydrogen fuel in a substantially completely gaseous phase, or in a substantially completely supercritical phase. The metering valve **90** provides the flow of fuel to the combustor **26** in a desired manner. More specifically, the metering valve **90** provides a desired volume of hydrogen fuel at, for example, a desired flow rate, to a fuel manifold that includes one or more fuel injectors that inject the hydrogen fuel into the combustor **26**. The fuel system **80** can include any components for supplying the fuel **67** from the fuel tank **82** to the combustor **26**, as desired.

[0048] The turbine engine **10** includes the steam system **100** in fluid communication with the one or more core exhaust nozzles **32** and the fan bypass nozzle **76**. The steam system **100** extracts steam from the combustion gases **66** as the combustion gases **66** flow through the steam system **100**, as detailed further below.

[0049] The turbine engine **10** depicted in FIG. **1** is by way of example only. In other exemplary embodiments, the turbine engine **10** may have any other suitable configuration. For example, in other exemplary embodiments, the fan **38** may be configured in any other suitable manner (e.g., as a fixed pitch fan) and further may be supported using any other suitable fan frame configuration. Moreover, in other exemplary embodiments, any other suitable number or configuration of compressors, turbines, shafts, or a combination thereof may be provided. In still other exemplary embodiments, aspects of the present disclosure may be incorporated into any other suitable turbine engine, such as, for example, turbofan engines, propfan engines, and/or turboprop engines.

[0050] FIG. **2** is a schematic diagram of the turbine engine **10** of FIG. **1** and the steam system **100** with a thermal transport system **200** according to an exemplary embodiment of the present disclosure. The turbine engine **10** is shown schematically in FIG. **2** and some components are not shown in FIG. **2**.
[0051] In at least one example embodiment, the steam system **100** includes a boiler **202**, a condenser **204**, a water separator **206**, a water pump **208**, and a steam turbine **210**. The boiler **202** is a heat exchanger that vaporizes liquid water from a water source to generate steam or water vapor, as detailed further below. The boiler **202** is thus a steam source. In particular, the boiler **202** is an exhaust gas-water heat exchanger. The boiler **202** is in fluid communication with the hot gas path **78** (FIG. **1**) and is positioned downstream of the LPT **30**. The boiler **202** is also in fluid communication with the water pump **208**, as detailed further below. The boiler **202** can include any type of boiler or heat exchanger for extracting heat from the combustion gases **66** and vaporizing liquid water into steam or water vapor as the liquid water and the combustion gases **66** flow through the boiler **202**.
[0052] The condenser **204** is a heat exchanger that further cools the combustion gases **66** as the combustion gases **66** flow through the condenser **204**, as detailed further below. In particular, the condenser **204** is an air-exhaust gas heat exchanger. The condenser **204** is in fluid communication with the boiler **202** and, in this embodiment, is positioned within the bypass airflow passage **56**. The condenser **204**, however, may be positioned elsewhere and thermally connected to other cooling sources, such as being thermally connected to the fuel **67** to transfer heat to the fuel **67**, particularly, when the fuel **67** is a cryogenic fuel such as hydrogen fuel. The condenser **204** can include any type of condenser for condensing water from the exhaust (e.g., the combustion gases **66**).
[0053] The water separator **206** is in fluid communication with the condenser **204** for receiving cooled exhaust (combustion gases **66**) having condensed water entrained therein. The water separator **206** is also in fluid communication with the one or more core exhaust nozzles **32** and with the water pump **208**. The water separator **206** includes any type of water separator for separating water from the exhaust. For example, the water separator **206** can include a cyclonic separator that uses vortex separation to separate the water from the air. In such embodiments, the water separator **206** generates a cyclonic flow within the water separator **206** to separate the water from the cooled exhaust. In FIG. **2**, the water separator **206** is schematically depicted as being in the nacelle **50**, but the water separator **206** could be located at other locations within the turbine engine **10**, such as, for example, radially inward of the nacelle **50**, closer to the turbomachine **16**. The water separator **206** may be driven to rotate by one of the engine shafts, such as the HP shaft **34** or the LP shaft **36**. As noted above, the boiler **202** receives liquid water from a water source to generate steam or water vapor. In the embodiment depicted in FIG. **2**, the condenser **204** and the water separator **206**, individually or collectively, are the water source for the boiler **202**.
[0054] The water pump **208** is in fluid communication with the water separator **206** and with the boiler **202**. The water pump **208** is in fluid communication with the condenser **204** via the water separator **206**. The water pump **208** may be any suitable pump, such as a centrifugal pump or a positive displacement pump. The water pump **208** directs the separated liquid water through the boiler **202** where it is converted back to steam. This steam is sent through the steam turbine **210**, then injected into working gas flow path **33**, such as into the combustor **26**.
[0055] In operation, the combustion gases **66**, also referred to as exhaust, flow from the LPT **30** into the boiler **202**. The combustion gases **66** transfer heat into the water **274** (e.g., in liquid form) within the boiler **202**, as detailed further below. The combustion gases **66** flow into the condenser **204**. The condenser **204** condenses the water **274** (e.g., in liquid form) from the combustion gases **66**. The bypass air **62** flows through the bypass airflow passage **56** and over or through the condenser **204** and extracts heat from the combustion gases **66**, cooling the combustion gases **66** and condensing the water **274** from the combustion gases **66**, to generate an exhaust-water mixture

270. The bypass air **62** is exhausted out of the turbine engine **10** through the fan bypass nozzle **76** to generate thrust, as detailed above. The condenser **204** thus may be positioned in bypass airflow passage **56**.

[0056] The exhaust-water mixture **270** flows into the water separator **206**. The water separator **206** separates the water **274** from the exhaust of the exhaust-water mixture **270** to generate separate exhaust **272** and the water **274**. The exhaust **272** is exhausted out of the turbine engine **10** through the one or more core exhaust nozzles **32** to generate thrust, as detailed above. The boiler **202**, the condenser **204**, and the water separator **206** thus also define a portion of the hot gas path **78** (see FIG. **1**) for routing the combustion gases **66**, the exhaust-water mixture **270**, and the exhaust **272** through the steam system **100** of the turbine engine **10**.

[0057] The water pump **208** pumps the water **274** (e.g., in liquid form) through one or more water lines (as indicated by the arrow for the water **274** in FIG. **2**) and the water **274** flows through the boiler **202**. As the water **274** flows through the boiler **202**, the combustion gases **66** flowing through the boiler **202** transfer heat into the water **274** to vaporize the water **274** and to generate the steam **276** (e.g., vapor). The steam turbine **210** includes one or more stages of steam turbine blades (not shown) and steam turbine stators (not shown). The steam **276** flows from the boiler **202** into the steam turbine **210**, through one or more steam lines (as indicated by the arrow for the steam **276** in FIG. **2**), causing the steam turbine blades of the steam turbine **210** to rotate, thereby generating additional work in an output shaft (e.g., one of the core shafts) connected to the turbine blades of the steam turbine **210**.

[0058] As noted above, the turbomachine **16** includes shafts, also referred to as core shafts, coupling various rotating components of the turbomachine **16** and other thrust producing components such as the fan **38**. In the turbomachine **16** shown in FIG. **1**, these core shafts include the HP shaft **34** and the LP shaft **36**. The steam turbine **210** is coupled to one of the core shafts of the turbomachine **16**, such as the HP shaft **34** or the LP shaft **36**. In the illustrated embodiment, the steam turbine **210** is coupled to the LP shaft **36**. As the steam **276** flows from the boiler **202** through the steam turbine **210**, the kinetic energy of this gas is converted by the steam turbine **210** into mechanical work in the LP shaft **36**. The reduced temperature steam (as steam **278**) exiting the steam turbine **210** is injected into the working gas flow path **33**, such as into the combustor **26**, upstream of the combustor **26**, or downstream of the combustor **26**. The steam **278** flows through one or more steam lines from the steam turbine **210** to the working gas flow path **33**. The steam **278** injected into the working gas flow path **33** adds mass flow to the core air **64** such that less core air **64** is needed to produce the same amount of work through the turbine section **27**. In this way, the steam system **100** extracts additional work from the heat in exhaust gas that would otherwise be wasted. The steam **278** injected into the working gas flow path **33** is in a range of about 20% to about 50% of the mass flow through the working gas flow path **33**.

[0059] The steam turbine **210** may have a pressure expansion ratio in a range of 2:1 to 6:1. The pressure expansion ratio is a ratio of the pressure at an inlet of the steam turbine **210** to the pressure at an exit of the steam turbine **210**. The steam turbine **210** may contribute approximately 25% of the power to the LP shaft **36** (or to the HP shaft **34**) when the steam system **100** recovers approximately 70% of the water **274** and converts the water **274** into the steam **276**. The steam turbine **210** has a pressure expansion ratio in a range of about 2:1 to about 6:1, the LPT **30** has a pressure expansion ratio in a range of about 4.5:1 to about 28:1, and the steam **278** contributes to about 20% to about 50% of the mass flow through the working gas flow path **33**. The steam turbine **210** expands the steam **276**, thereby reducing the energy of the steam **278** exiting the steam turbine **210** and reducing the temperature of the steam **278** to approximately a temperature of the compressed air **65** (see FIG. **1**) that is discharged from the HPC **24**. Such a configuration enables the steam **278** to reduce hot spots in the combustor **26** from the combustion of the fuel (e.g., in particular when the fuel is supercritical hydrogen or gaseous hydrogen).

[0060] The steam **278** injected into the working gas flow path **33** also enables the HPT **28** to have a greater energy output with fewer stages of the HPT **28** as compared to HPTs without the benefit of the present disclosure. For example, the additional mass flow from the steam **278** through the turbine section **27** helps to produce a greater energy output. In this way, the HPT **28** may only have one stage capable of sustainably driving a higher number of stages of the HPC **24** (e.g., 10, 11, or 12 stages of the HPC **24**) due to the higher mass flow (resulting from the steam injection) exiting the combustor **26**. The steam **278** that is injected into the working gas flow path **33** enables the HPT **28** to have only one stage that drives the plurality of stages of the HPC **24** without reducing an amount of work that the HPT **28** produces as compared to HPTs without the benefit of the present disclosure, while also reducing a weight of the HPT **28** and increasing an efficiency of the HPT **28**, as compared to HPTs without the benefit of the present disclosure.

[0061] With less core air **64** (see FIG. **1**) needed due to the added mass flow from the steam **276**, the compression ratio of the HPC **24** may be increased as compared to HPCs without the benefit of the present disclosure. In this way, the HPC **24** has a compression ratio greater than about 20:1. In some embodiments, the compression ratio of the HPC **24** is in a range of about 20:1 to about 40:1. Thus, the compression ratio of the HPC **24** is increased, thereby increasing the thermal efficiency of the turbine engine **10** as compared to HPCs and turbine engines without the benefit of the present disclosure. Further, the HPC **24** may have a reduced throat area due to the added mass flow in the turbomachine **16** provided by the steam **276**, **278** injected into the turbomachine **16**. Accordingly, the HPC **24** has a reduced size (e.g., outer diameter) and a reduced weight, as compared to turbine engines without the benefit of the present disclosure.

[0062] In some embodiments, the HPC stator vanes of at least two stages of the HPC **24** are variable stator vanes that are controlled to be pitched about a pitch axis to vary a pitch of the HPC stator vanes. In some embodiments, the HPC **24** includes one or more compressor bleed valves that are controlled to be opened to bleed a portion of the compressed air **65** (see FIG. **1**) from the HPC **24**. The one or more compressor bleed valves are preferably positioned between a fourth stage of the HPC **24** and a last stage of the HPC **24**. The HPC stator vanes that are variable stator vanes, and the one or more compressor bleed valves help to balance the air flow (e.g., the compressed air **65**) through the stages of the HPC **24**. Such a balance, in combination with the steam **278** injected into the working gas flow path **33**, enables the number of stages of the HPC **24** to include ten to twelve stages for compression ratios to be greater than about 20:1, and preferably in a range of about 20:1 to about 40:1.

[0063] The additional work that is extracted by the steam system **100** and the steam **278** injected into the working gas flow path **33** enables a size of the turbomachine **16** (FIG. **1**) to be reduced, thereby increasing the bypass ratio of the turbine engine **10**, as compared to turbine engines without the benefit of the present disclosure. In this way, the turbine engine **10** has a bypass ratio greater than about 18:1, preferably, in a range of about 18:1 to about 100:1, more preferably, in a range of about 25:1 to about 85:1, and, most preferably, in a range of about 28:1 to about 70:1. In this way, the steam system **100** can enable an increased bypass ratio in which the turbine engine **10** can move a larger mass of air through the bypass, reducing the pressure ratio of the fan **38** and increasing the efficiency of the turbine engine **10** as compared to turbine engines without the benefit of the present disclosure.

[0064] The turbine engine **10** may also include an engine controller **220**. The engine controller **220** is configured to operate various aspects of the turbine engine **10**, including, in this embodiment, the water pump **208**, a fuel bypass valve **214**, a selector valve **228**, a first heater **232**, a second heater **234**, and a third heater **236**. The engine controller **220** may be a Full Authority Digital Engine Control (FADEC). In this embodiment, the engine controller **220** is a computing device having one or more processors **222** and one or more memories **224**. The processor **222** can be any suitable processing device, including, but not limited to, a microprocessor, a microcontroller, an integrated circuit, a logic device, a programmable logic controller (PLC), an application-specific integrated circuit (ASIC), and/or a Field Programmable Gate Array (FPGA). The memory **224** can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, a computer-readable non-volatile medium (e.g., a flash memory), a RAM, a ROM, hard drives, flash drives, and/or other memory devices.

[0065] The memory **224** can store information accessible by the processor **222**, including computer-readable instructions that can be executed by the processor **222**. The instructions can be any set of instructions or a sequence of instructions that, when executed by the processor **222**, causes the processor **222** and the engine controller **220** to perform operations. In some embodiments, the instructions can be executed by the processor **222** to cause the processor **222** to complete any of the operations and functions for which the engine controller **220** is configured, as will be described further below. The instructions can be software written in any suitable programming language, or can be implemented in hardware. Additionally, and/or alternatively, the instructions can be executed in logically and/or virtually separate threads on the processor **222**. The memory **224** can further store data that can be accessed by the processor **222**.

[0066] The technology discussed herein makes reference to computer-based systems and actions taken by, and information sent to and from, computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between components and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.

[0067] The fuel **67** (FIG. **1**) may be heated prior to injecting the fuel **67** into the combustor **26** for various reasons. Some fuels, such as hydrocarbon-based fuels may be heated, for example, to prevent ice formation or for performance benefits. Other fuels, such as cryogenic fuels, may be heated and vaporized (converted from a liquid state in which the cryogenic fuel is stored to a vapor state for combustion), prior to being injected into the combustor **26**. As noted above, the embodiments discussed herein utilize a thermal transport system to transfer heat from the turbine engine **10** and to use that heat to help heat the fuel **67**. More specifically, in the embodiments discussed herein, the water **274** is used as the heat transfer fluid to transfer heat from the steam system **100** to the fuel **67** in the fuel system **80**. The following discussion makes reference to cryogenic fuel and, more specifically, hydrogen fuel, but the thermal transport systems discussed herein may be applicable to other fuel systems.

[0068] The fuel tank **82** is configured to hold the hydrogen fuel at least partially within the liquid phase and is configured to provide hydrogen fuel to the fuel delivery assembly **84** substantially completely in the liquid phase, such as completely in the liquid phase. The fuel tank **82** has a fixed volume and contains a volume of the hydrogen fuel in the liquid phase (e.g., liquid hydrogen fuel). As the fuel tank **82** provides hydrogen fuel to the fuel delivery assembly **84** substantially completely in the liquid phase, the volume of the liquid hydrogen fuel in the fuel tank **82** decreases and the remaining volume in the fuel tank **82** is made up by, for example, hydrogen substantially completely in the gaseous phase (gaseous hydrogen). As used herein, the term “substantially completely” is used to describe a phase of the hydrogen fuel refers to at least 99% by mass of the described portion of the hydrogen fuel being in the stated phase, such as at least 97.5%, such as at least 95%, such as at least 92.5%, such as at least 90%, such as at least 85%, or such as at least 75% by mass of the described portion of the hydrogen fuel being in the stated phase.

[0069] To store the hydrogen fuel substantially completely in the liquid phase, the hydrogen fuel is stored in the fuel tank **82** at very low (cryogenic) temperatures, and, thus, the fuel tank **82** may also be referred to herein as a cryogenic fuel tank. For example, the hydrogen fuel may be stored in the fuel tank **82** at about −253 degrees Celsius (twenty Kelvin) or less at atmospheric pressure, or at other temperatures and pressures to maintain the hydrogen fuel substantially completely in the liquid phase. In some embodiments, the hydrogen fuel may be stored in the fuel tank **82** at

temperatures from about −259 degrees Celsius (about fourteen Kelvin) to about −243 **0** degrees Celsius (about thirty Kelvin), and, more preferably, from about −253 degrees Celsius (about twenty Kelvin) to about −243 degrees Celsius (about thirty Kelvin). To store the hydrogen fuel in the liquid phase, the fuel tank **82** stores and maintains the hydrogen cryogenically and may be a cryostat. The fuel tank **82** may thus be, for example, a dual wall tank, including an inner vessel (e.g., an inner cryogenic liquid tank) and an outer vessel (e.g., a vacuum vessel). The inner vessel may be positioned within the outer vessel with a gap formed between the inner vessel and the outer vessel. To provide thermal isolation for the inner vessel, the gap may be under vacuum. The gap may include a void space or be an entirely void space, but, alternatively, the gap may include multi-layer insulation (MLI), such as aluminized polyester films (e.g., aluminized Mylar®), for example.
[0070] The turbine engine **10** shown in FIG. **2** includes the thermal transport system **200** that may be used to transfer heat from the steam system **100** to the fuel system **80**. The thermal transport system **200** includes a fuel heat exchanger **211** fluidly connected to the fuel delivery assembly **84**. The fuel heat exchanger **211** of this embodiment is a fuel/water heat exchanger, and as the fuel **67** flows from the fuel tank **82** to the combustor **26**, the fuel **67** absorbs (receives) heat from the water **274** of the steam system **100** and is heated. In some example embodiments, the fuel heat exchanger **211** may be one of the vaporizers **88** discussed above. Additionally, the fuel heat exchanger **211** may be in addition to the vaporizers **88**. The fuel heat exchanger **211** may be any suitable heat exchanger, including, for example, a plate heat exchanger or a tubular heat exchanger, such as a tube and shell heat exchanger. The fuel heat exchanger **211** thus includes a fluid flow path for the heat transfer fluid.
[0071] There may be instances when heating the fuel **67** with the water **274** is not desirable or when removing heat from the water **274** is not desirable. Such conditions include startup, as discussed further below, when the water **274** is not sufficiently hot enough to flow through the fuel heat exchanger **211** and to heat the fuel **67**. Accordingly, the fuel system **80** of this embodiment includes a fuel bypass line **215** (e.g., a fuel bypass flow path) that fluidly connects a portion of the fuel delivery assembly **84** upstream of the fuel heat exchanger **211** with a portion of the fuel delivery assembly **84** downstream of the fuel heat exchanger **211**, thus, bypassing the fuel heat exchanger **211**. The fuel system **80** is, thus, selectively operable to redirect the fuel **67**, or a portion thereof, and to bypass the fuel heat exchanger **211**. The fuel bypass line **215** includes a fuel bypass valve **214** located in the fuel bypass line **215** and the fuel delivery assembly **84**. The fuel bypass valve **214** is operable to open and to direct the fuel **67** through the fuel bypass line **215**, bypassing the fuel heat exchanger **211**, and, thus, the fuel bypass valve **214** selectively operates the fuel system **80** to bypass the fuel heat exchanger **211**. The fuel bypass valve **214** may be any suitable valve including a three-way valve. The fuel bypass valve **214** may also be a flow control valve (e.g., a proportional control valve) that directs a portion of the fuel **67** and or controls the flow of the fuel **67** through the fuel heat exchanger **211** and the fuel bypass line **215**. The fuel bypass valve **214** may be any suitable valve including, for example, an electrically operable valve, a hydraulically operable valve, or a pneumatically operable valve. When the fuel bypass valve **214** is hydraulically operable, the hydraulic fluid may be suitable fluids of the turbine engine **10** including, for example, the fuel **67**, lubrication oil, and the like.
[0072] The thermal transport system **200** includes a heat transfer loop **225** that thermally couples the condenser **204** with the fuel heat exchanger **211** to transfer heat from the condenser **204** with the fuel heat exchanger **211**. More specifically, the water **274** flows through the heat transfer loop **225** to transfer heat from the condenser **204** to the fuel heat exchanger **211**. The heat transfer loop **225** includes a supply line **230**. The supply line **230** is fluidly connected to the line conveying the water **274** between the water pump **208** and the boiler **202**. Thus, the supply line **230** is fluidly connected downstream of the water pump **208** and upstream of the boiler **202**. As noted above, the water pump **208** directs the separated liquid water **274** through the boiler **202**, and, with the location of the water pump **208** relative to the supply line **230**, the water pump **208** may also be

used to direct the flow of the water **274** through the thermal transport system **200**. The water **274** thus flows from the water pump **208** through the supply line **230** to the fuel heat exchanger **211**.

[0073] Although the exhaust-water mixture **270** has been cooled to a temperature low enough to condense water from the combustion gases **66**, the temperature of the water **274** may still be relatively high, such as from about one hundred degrees Fahrenheit (100° F.) (about thirty-eight degrees Celsius (38° C.)) to about two hundred twelve degrees Fahrenheit (212° F.) (about one hundred degrees Celsius (100° C.)) or from about one hundred fifty degrees Fahrenheit (150° F.) (about sixty-six degrees Celsius (66° C.)) to about two hundred twelve degrees Fahrenheit (212° F.) (about one hundred degrees Celsius (100° C.)). Such temperatures are still higher than the temperature of the fuel **67**, particularly, when the fuel is a cryogenic liquid fuel, such as hydrogen fuel, stored in the fuel tank **82** at cryogenic temperatures. Thus, as the water **274** flows through the fuel heat exchanger **211**, the heat from the water **274** is transferred from the water **274** to the fuel **67**, and the fuel **67** absorbs the heat from the water **274**. The fuel **67** is thus heated and may be vaporized by the heat from the water **274** and the water **274** is cooled.

[0074] The water **274** flows from the fuel heat exchanger **211** back to the boiler **202** where the water **274** can be heated to form the steam **276** in the manner discussed above. The thermal transport system **200** thus includes a return line **226** through which the water **274** flows from the fuel heat exchanger **211** and into the boiler **202**. In FIG. **2**, the return line **226** is shown as being fluidly connected to the boiler **202**, but, alternatively, the return line **226** may be fluidly connected to a water line upstream of the boiler **202**, such as a waterline directly connecting the water pump **208** with the boiler **202**.

[0075] The water **274**, having been cooled by the fuel **67**, may be preheated before flowing into the boiler **202**. A preheat heat exchanger may be located in the return line **226** to preheat the water **274**. In this embodiment, the preheat heat exchanger is the condenser **204**. The thermal transport system **200** includes an intermediate return line **226** fluidly connecting the fuel heat exchanger **211** to a flow passage within the condenser **204**, and the return line **226** fluidly connects the flow passage of the condenser **204** to the boiler **202**. The water **274** may thus be preheated by the combustion gases **66** flowing through the condenser **204** before being introduced into the boiler **202**.

[0076] The heat transfer loop **225** may be selectively operable such that the heat transfer loop **225** may be isolated or such that only a portion of the water **274** flows through the heat transfer loop **225**. The heat transfer loop **225** includes a selector valve **228** located in the water line between the water pump **208** and the boiler **202**. The selector valve **228** is operable to open and to direct the water **274** through the supply line **230** and to the fuel heat exchanger **211**. The selector valve **228** may be any suitable valve including a three-way valve. The selector valve **228** may also be a flow control valve (e.g., a proportional control valve) that directs a portion of the water **274** and or controls the flow of the water **274** through the heat transfer loop **225**. As with the fuel bypass valve **214** discussed above, the selector valve **228** may be an electrically operable valve, a hydraulically operable valve, or a pneumatically operable valve.

[0077] The water **274** is used as the heat transfer medium of the heat transfer loop **225** of this embodiment. To avoid freezing the water **274**, the thermal transport system **200** includes a plurality of heaters, including a first heater **232**, a second heater **234**, and a third heater **236**. Any suitable heater may be used including, for example, one or more electrical resistance heaters, a catalytic heater, or a burner. In some environments, startup of the turbine engine **10** may occur at cold temperatures (e.g., as low as negative forty degrees Fahrenheit (−40° F.) (negative forty degrees Celsius (−40° C.))). At startup, the turbine engine **10** and, more specifically, the combustor **26**, has not begun to operate and, thus, the components in the hot gas path **78** (FIG. **1**) are closer to ambient conditions than their operational temperatures. With the components of the steam system **100** and/or the thermal transport system **200** at these cold temperatures, the water **274** within the steam system **100** and the thermal transport system **200** may freeze when the water **274** comes into contact with these components. To avoid such freezing, the heaters **232**, **234**, **236** are used to increase the temperature of these components above the freezing point of water (thirty-two degrees Fahrenheit (32° F.) (zero degrees Celsius (0° C.))), such as from thirty-five degrees Fahrenheit (35° F.) (one degree Celsius (1° C.)) to fifty degrees Fahrenheit (50° F.) (ten degrees Celsius (10° C.)). The first heater **232** may be located in the fuel heat exchanger **211** and used to raise the temperature of the fuel heat exchanger **211**. The second heater **234** may be located in the water separator **206** and used to raise the temperature of the water separator **206**. The third heater **236** may be located in the condenser **204** and used to raise the temperature of the condenser **204**.

[0078] One or more temperature sensors (e.g., a first temperature sensor **240**, a second temperature sensor **242**, a third temperature sensor **244**, and a fourth temperature sensor **246**) may be used with the heaters **232**, **234**, **236** to provide input to operate the heaters **232**, **234**, **236** to heat the components as discussed above. The second temperature sensor **242** may be located within the fuel heat exchanger **211** and used to detect the temperature of the fuel heat exchanger **211**. The second temperature sensor **242** may be communicatively coupled to the first heater **232** and, thus, provides input to operate the first heater **232** based on the temperatures sensed by the second temperature sensor **242**. Similarly, the fourth temperature sensor **246** is located within the condenser **204** and may be used to detect the temperature of the condenser **204**. The third heater **236** may be operated to increase the temperature of the condenser **204** based on the temperature sensed by the fourth temperature sensor **246**. The first heater **232** and the third heater **236** may thus be operated to increase the temperature of the fuel heat exchanger **211** and the third heater **236**, respectively, to increase the temperatures of these components, as discussed above.

[0079] When the temperatures of the fuel heat exchanger **211** and the condenser **204** have increased sufficiently, the water **274** may be circulated through the water line connecting the water separator **206** to the boiler **202** and the heat transfer loop **225**. The water **274** may be initially heated within the water separator **206** by the second heater **234**. The first temperature sensor **240** may be located in the supply line **230** to determine the temperature of the supply line **230** and the water **274** flowing therein. Similarly, the third temperature sensor **244** may be located in a water line positioned downstream of the water separator **206** and upstream of the boiler **202** to determine the temperature of the water line and the water **274** flowing therein. The second heater **234** positioned in the water separator **206** may be used to increase the temperature of the water separator **206** and also the water **274** before flowing through the steam system **100** and the thermal transport system **200**. The first temperature sensor **240** and the third temperature sensor **244** may thus be used to monitor the temperature of the water **274** and to provide input to operate the heaters **232**, **234**, **236** based on these temperatures.

[0080] When the fuel **67** is a cryogenic fuel, the fuel **67** may be directed through the fuel bypass line **215** during startup operations to avoid cooling the fuel heat exchanger **211** and negating the effects of the first heater **232**. As the water **274** increases in temperature, as detected by, for example, the first temperature sensor **240**, the fuel bypass valve **214** may be operated to slowly introduce the fuel **67** into the fuel heat exchanger **211**.

[0081] A method of starting-up the turbine engine **10** may start with the steam injection system off due to the potential for sub-freezing temperatures. The fuel bypass valve **214** may be positioned to bypass the fuel heat exchanger **211** and have fuel **67** flow through the fuel bypass line **215**. The selector valve **228** is also positioned to direct the water **274** to the boiler **202** and to bypass the heat transfer loop **225**. The turbine engine **10** is started with the water pump **208** off. The turbine engine **10** is brought up to idle using, for example, a starter motor. Using the second temperature sensor **242**, the temperature of the fuel heat exchanger **211** is measured. If the temperature of the fuel heat exchanger **211** is not above a heater threshold temperature, the first heater **232** is used to heat the fuel heat exchanger **211** to the temperatures discussed above. Preferably, the heater threshold temperature is greater than the freezing temperature of water (e.g., greater than thirty-two degrees Fahrenheit (32° F.) (zero degrees Celsius (0° C.))), such as from thirty-five degrees Fahrenheit (35° F.) (one degree Celsius (1° C.)) to fifty degrees Fahrenheit (50° F.) (ten degrees Celsius (10° C.)). If (or once) the temperature of the fuel heat exchanger **211** reaches the heater threshold temperature, the water pump **208** and/or the selector valve **228** is modulated (controlled) to send at least some of the water **274** through the heat transfer loop **225**.

[0082] Once the water **274**, flowing through the fuel heat exchanger **211**, is above a minimum threshold as measured by the first temperature sensor **240**, the fuel bypass valve **214** is used to modulate (e.g., control) the flow of fuel **67** into the fuel heat exchanger **211**. The fuel **67** may be modulated using the fuel bypass valve **214** to maintain the water **274** above the minimum threshold temperature. The water **274** is preferably at least about eighty degrees Fahrenheit (80° F.) (about twenty-six degrees Celsius (26° C.)) before the fuel **67** is introduced into the fuel heat exchanger to prevent instant freezing of the water **274**, and thus the minimum threshold may be, for example, about eighty degrees Fahrenheit (80° F.) (about twenty-six degrees Celsius (26° C.)) or more, such as, for example, from about eighty degrees Fahrenheit (80° F.) (about twenty-six degrees Celsius (26° C.)) to about one hundred sixty degrees Fahrenheit (160° F.) (about seventy-one degrees Celsius (71° C.)). A temperature at the inlet to the fuel heat exchanger **211** of about one hundred sixty degrees Fahrenheit (160° F.) (about seventy-one degrees Celsius (71° C.)) or less minimizes the potential foil boiling at altitude. As the temperature of the water **274** increases due to operation of the turbine engine **10**, the selector valve **228** and/or the water pump **208** may be controlled to increase the flow of the water **274** through the thermal transport system **200** to a maximum position. The fuel bypass valve **214** may be used to modulate the fuel **67** flowing into the fuel heat exchanger **211** as the flow of water **274** flowing through the heat transfer loop **225** is increased. The flow of the water **274** and the flow of the fuel **67** are increased until the turbine engine **10** reaches a full idle operation.

[0083] Once the combustor **26** is producing the combustion gases **66**, the condenser **204** and the water separator **206** have the hot combustion gases **66** flowing therethrough, providing heat to these components. During the start-up sequence, particularly for start-up when the turbine engine **10** has been exposed to sub-freezing temperatures while shutdown, the second heater **234** and the third heater **236** may be used to heat the water separator **206** and the condenser **204**, respectively, to temperatures above the freezing point of water (about thirty-two degrees Fahrenheit (32° F.)), such as from about thirty-five degrees Fahrenheit (35° F.) (about one degree Celsius (1° C.)) to about fifty degrees Fahrenheit (50° F.) (about ten degrees Celsius (10° C.)). The method of starting-up the turbine engine **10** thus may also include using the fourth temperature sensor **246** to measure the temperature of the condenser **204**. If the temperature of the condenser **204** is not above a heater threshold temperature (discussed above), the fourth temperature sensor **246** is used to heat the condenser **204** to the temperatures discussed above. If (or once) the temperature of the condenser **204** reaches the heater threshold temperature, the turbine engine **10** can be started, such as by igniting the fuel **67** in the combustor **26**. Once the combustor ignition has occurred, the second heater **234** and the third heater **236** may be turned off. In some embodiments, the third temperature sensor **244**, positioned in the water line fluidly connecting the water separator **206** with the water pump **208**, can be used to control the second heater **234** during the startup sequence and determine when to turn off the second heater **234**.

[0084] A method of shutting down the turbine engine **10** may include decelerating the turbine engine **10** to ground idle thrust. The fuel bypass valve **214** is positioned to bypass the fuel **67** around the fuel heat exchanger **211** and through the fuel bypass line **215**. The selector valve **228** is positioned to direct the water **274** directly to the boiler **202**, bypassing the heat transfer loop **225**. The water pump **208** is shut off. The combustion gases **66** continue to flow through the hot gas path **78** with the water pump **208** shut off allowing the condenser **204** and the water separator **206** to be dried out at temperatures above freezing. Once dry, the pump **86** that pumps the fuel **67** from the fuel tank **82**, through the fuel delivery assembly **84**, can be shut off to shut down the turbine engine **10**.

[0085] As noted above, the engine controller **220** is configured to operate various aspects of the turbine engine **10**, including, in this embodiment, the components described in the methods above. Accordingly, in some embodiments, the engine controller **220** is configured to execute the steps of the method discussed above. More specifically, for example, the processor **222** may execute a sequence of instructions stored on the memory **224** to operate the turbine engine **10** in the manner described above.

[0086] FIG. **3** is a schematic diagram of a vapor generation system of the turbine engine **10** of FIGS. **1-2** according to an exemplary embodiment of the present disclosure.

[0087] In at least one example embodiment, the steam system **100** of the turbine engine **10** includes a vapor generation apparatus or system, such as a heat exchanger assembly **300**. For example, the heat exchanger assembly **300** may be incorporated into the steam system **100** as the boiler **202**, described above with respect to FIG. **2**. The heat exchanger assembly **300** may include a first fluid channel **305**, a plurality of first fluid passageways **310**, and a plurality of second fluid passageways **315**. Each of the first fluid channel **305**, the plurality of first fluid passageways **310**, and the plurality of second fluid passageways **315** extend between a first end **301** and a second end **302** opposite the first end **301** of the heat exchanger assembly **300**. Additionally, each of the first fluid channel **305**, the plurality of first fluid passageways **310**, and the plurality of second fluid passageways **315** may include a plurality of tubes or conduits extending between the first end **301** and the second end **302**.

[0088] In at least one example embodiment, the first fluid channel **305**, the plurality of first fluid passageways **310**, and the plurality of second fluid passageways **315** are arranged in a stacked relationship. For example, the plurality of first fluid passageways **310** may be between the first fluid channel **305** and the plurality of second fluid passageways **315**. Moreover, the first fluid channel **305** may be adjacent a first side **321** of the heat exchanger assembly **300** and the plurality of second fluid passageways **315** may be adjacent a second side **322** opposite the first side **321** of the heat exchanger assembly **300**.

[0089] In at least one example embodiment, the heat exchanger assembly **300** includes a fluid chamber **320** configured to store a first fluid. The first fluid may be a liquid, such as water. The fluid chamber **320** may be adjacent the second end **302** of the heat exchanger assembly **300** and in fluid communication with the first fluid channel **305** and the plurality of first fluid passageways **310**, as will be described in greater detail below. Moreover, the heat exchanger assembly **300** may include an inlet nozzle **323** in fluid communication with the fluid chamber **320**. The inlet nozzle **323** may function as an ejector by creating a low static pressure region near the inlet nozzle **323** in order to draw liquid from the first fluid channel **305** to the fluid chamber **320**. Additionally, the inlet nozzle **323** may be configured to deliver the first fluid, such the water **274**, to the fluid chamber **320** of the heat exchanger assembly **300**, as discussed above with respect to FIG. **2**.

[0090] In at least one example embodiment, the plurality of first fluid passageways **310** receive the first fluid from the fluid chamber **320**. The plurality of first fluid passageways **310** are configured to heat the first fluid as it travels through the plurality of first fluid passageways **310** from the second end **302** towards the first end **301**. For example, the heat exchanger assembly **300** may be in fluid communication with the working gas flow path **33** and configured to receive the combustion gases **66**. The combustion gases **66** flow through the heat exchanger assembly **300** and transfer heat to the first fluid, such as the water **274**, flowing through the plurality of first fluid passageways **310** to generate a second fluid, such as the steam **276**, as discussed above with respect to FIG. **2**. Moreover, the plurality of first fluid passageways **310** are configured to heat the first fluid (the water **274**) to generate a gas-vapor mixture. For example, the gas-vapor mixture includes the water **274** and the steam **276**. In at least one example embodiment, the gas-vapor mixture may be greater than or equal to about 90% vapor, such as the steam **276**, by mass.

[0091] In at least one example embodiment, the heat exchanger assembly **300** includes a separator **325** adjacent the first end **301**. The separator **325** may be in fluid communication with the first fluid channel **305**, the plurality of first fluid passageways **310**, and the plurality of second fluid passageways **315**. The separator **325** is configured to separate the second fluid from the first fluid. For example, the separator **325** may include an inertial separator, as will be discussed in greater detail with respect to FIGS. **5**A-**5**B below. The second fluid may include vapor or steam, such as the steam **276**.

[0092] The second fluid and any remaining amount of the first fluid enter the separator **325** from the plurality of first fluid passageways **310**. The remaining amount of the first fluid is delivered from the separator **325** to the first fluid channel **305**. The first fluid channel **305** is configured to direct the first fluid to the fluid chamber **320**, where it may be recirculated through the heat exchanger assembly **300**. In at least one example embodiment, the first fluid channel **305** includes wicking material. The wicking material may be disposed at least partially along a length of the first fluid channel **305**. The wicking material is configured to guide the first fluid within the first fluid channel **305** to the fluid chamber **320**. In at least one example embodiment, the wicking material includes a mesh, sintered metal, or other porous materials.

[0093] The second fluid exits the separator **325** and travels to the plurality of second fluid passageways **315**. The heat exchanger assembly **300** may include a second fluid inlet **330** adjacent the first end **301**. The second fluid inlet **330** may be configured to receive the second fluid and deliver the second fluid to the plurality of second fluid passageways **315**. The second fluid may be heated within the plurality of second fluid passageways **315**. For example, the heat exchanger assembly **300** may be in fluid communication with the working gas flow path **33** and receive the combustion gases **66**. The combustion gases **66** flow through the heat exchanger assembly **300** and transfer heat to the second fluid within the plurality of second fluid passageways **315**. The second fluid travels through the plurality of second fluid passageways **315** to a second fluid outlet **335** adjacent the second end **302**. The second fluid, such as the steam **276**, may be directed from the second fluid outlet **335** of the heat exchanger assembly **300** to the steam turbine **210**, as discussed above with respect to FIG. **2**.

[0094] FIG. **4**A is a perspective, detailed view of the separator **325** of the heat exchanger assembly **300** of FIG. **3** according to an exemplary embodiment of the present disclosure. FIG. **4**B is a cross-sectional view of the separator **325** of the heat exchanger assembly **300** of FIG. **4**A according to an exemplary embodiment of the present disclosure. FIG. **4**C is a detailed, cross-sectional view of the separator **325** of the heat exchanger assembly **300** of FIG. **4**B according to an exemplary embodiment of the present disclosure.

[0095] In at least one example embodiment, the separator **325** of the heat exchanger assembly **300** includes a plurality of separation passageways **400** extending between a first separator side **401** and a second separator side **402** opposite the first separator side **401**. For example, the first separator side **401** may be adjacent the first end **301** of the heat exchanger assembly **300** and the second separator side **402** may be adjacent the plurality of first fluid passageways **310**. The plurality of separation passageways **400** are in fluid communication with the plurality of first fluid passageways **310**. The plurality of separation passageways **400** may include a plurality of tubes or conduits extending between the first separator side **401** and the second separator side **402**. In at least one example embodiment, a number of the plurality of separation passageways **400** may be the same as a number of the plurality of first fluid passageways **310**. In some additional example embodiments, the plurality of separation passageways **400** may be integral with the plurality of first fluid passageways **310**.

[0096] In at least one example embodiment, the separator **325** of the heat exchanger assembly **300** includes a separation chamber **405** adjacent the first separator side **401**. The separation chamber **405** is in fluid communication with the plurality of separation passageways **400**. For example, as shown in FIGS. **4**B-**4**C, each of the plurality of separation passageways **400** define a plurality of fluid openings **408** about a periphery of the plurality of separation passageways **400** adjacent the first separator side **401**. The plurality of fluid openings **408** configured to fluidly couple the plurality of separation passageways **400** with the separation chamber **405**. The plurality of fluid openings **408** may include a plurality of perforations or slits disposed in a periphery of each of the plurality of separation passageways **400**.

[0097] With reference to FIGS. **4**B-**4**C, the separation chamber **405** is also configured to be in fluid communication with the first fluid channel **305**. For example, the first fluid is configured to exit each of the plurality of separation passageways **400** through the plurality of fluid openings **408** and flow into the first fluid channel **305** from the separation chamber **405**. Moreover, a periphery of the first fluid channel **305** may define at least one inlet opening, such as a first fluid inlet **410**, adjacent the first end **301** and in fluid communication with the separation chamber **405**.

[0098] In at least one example embodiment, the heat exchanger assembly **300** includes a second fluid channel **415**. The second fluid channel **415** may extend vertically adjacent the first end **301** of the heat exchanger assembly **300**. For example, the second fluid channel **415** may be adjacent the first end **301** of the first fluid channel **305**, the separator **325**, and the plurality of second fluid passageways **315**. The second fluid channel **415** is in fluid communication with the separator **325** and the plurality of second fluid passageways **315**. More specifically, the second fluid channel **415** is in fluid communication with the plurality of separation passageways **400** and the plurality of second fluid passageways **315** to provide a fluid pathway for the second fluid to travel from the plurality of separation passageways **400** to the plurality of second fluid passageways **315**. Additionally, the second fluid channel **415** is fluidly isolated from the separation chamber **405** and the first fluid channel **305**. Such fluid isolation prevents the first fluid from entering the second fluid channel **415**, and thereby the plurality of second fluid passageways **315**.

[0099] In at least one example embodiment, the separator **325** is configured to separate the first fluid and the second fluid entering the separator **325** from the plurality of first fluid passageways **310**. For example, each of the plurality of separation passageways **400** define a first fluid pathway **420** and a second fluid pathway **425**. With reference to FIG. **4**C, the first fluid enters the plurality of separation passageways **400** adjacent the second separator side **402** from the plurality of first fluid passageways **310**. The first fluid pathway **420** of the first fluid extends from the second separator side **402** towards the first separator side **401**, through the plurality of fluid openings **408**, through the separation chamber **405**, and into the first fluid channel **305** via the first fluid inlet **410**. The first fluid may be returned to the fluid chamber **320** via the first fluid channel **305**, as discussed above with respect to FIG. **3**.

[0100] With reference to FIG. **4**B, the second fluid enters the plurality of separation passageways **400** adjacent the second separator side **402** from the plurality of first fluid passageways **310**. The plurality of separation passageways **400** are configured to separate the second fluid from the first fluid, as will be described in greater detail with respect to FIGS. **5**A-**5**B, below. The second fluid flows along the second fluid pathway **425** from the second separator side **402** towards the first separator side **401**, exits the plurality of separation passageways **400** through a second fluid outlet **430** adjacent the first separator side **401**, flows into the second fluid channel **415**, and enters the plurality of second fluid passageways **315**. The second fluid may be delivered to the second fluid outlet **335** via the plurality of second fluid passageways **315**, as described above with respect to FIG. **3**.

[0101] FIG. **5**A is a perspective view of one of the plurality of separation passageways **400** of the separator **325** of FIGS. **4**A-**4**C according to an exemplary embodiment of the present disclosure. FIG. **5**B is a rear, perspective view of the one of the plurality of separation passageways **400** of FIG. **5**A according to an exemplary embodiment of the present disclosure.

[0102] In at least one example embodiment, each of the plurality of separation passageways **400** include a base portion **500** adjacent the second separator side **402** and a tapered surface or a tapered portion **505** adjacent the first separator side **401**. For example, a diameter the base portion **500** may be constant from the second separator side **402** to the tapered portion **505** and a diameter of the tapered portion **505** decreases from the base portion **500** to the first separator side **401**. The tapered portion **505** collects water vapor, such as the steam **276**. A periphery of the tapered portion **505** of each of the plurality of separation passageways **400** may define the plurality of fluid openings **408**. Additionally, the tapered portion **505** may define the second fluid outlet **430** adjacent the first separator side **401**. The second fluid outlet **430** is in fluid communication with the plurality of separation passageways **400**.

[0103] In at least one example embodiment, the separator **325** includes an inertial separator. In such embodiments, a plurality of vanes **510** are disposed in each of the plurality of separation passageways **400** adjacent the second separator side **402**. For example, the plurality of vanes **510** may be disposed circumferentially about a separator hub **515**. The separator hub **515** and the plurality of vanes **510** may define a swirler **518** disposed in each of the plurality of separation passageways **400** adjacent the second separator side **402**. The plurality of vanes **510** are configured to add swirl to the first fluid and the second fluid entering the plurality of separation passageways **400** from the plurality of first fluid passageways **310** such that the first fluid and the second fluid are separated. The plurality of vanes **510** also facilitate smaller liquid droplets coalescing into larger liquid droplets, which may improve the liquid separation process within the separator **325**.

[0104] In operation, the first fluid, such as the water **274**, enters the fluid chamber **320** and flows through the plurality of first fluid passageways **310**. With reference to FIG. **2**, the fluid chamber **320** may receive the water **274** from a water source, such as one or both of the condenser **204** and the water separator **206**. Within the plurality of first fluid passageways **310**, the first fluid undergoes convective boiling that converts a portion of the first fluid (the water **274**) to the second fluid (the steam **276**). More specifically, only a portion of the first fluid is converted to the second fluid. For example, the boiling process is incomplete because not all the water **274** within the plurality of first fluid passageways **310** contacts the interior surface of the plurality of passageways to evaporate into vapor and form the steam **276**. Accordingly, the flow entering the second separator side **402** of the separator **325** is a gas-liquid mixture.

[0105] The separator **325** separates the second fluid (the steam **276**) from the first fluid (the water **274**), as described above. The steam **276** flows to the plurality of second fluid passageways **315**. Within the plurality of second fluid passageways **315**, the steam **276** is further heated. For example, energy transferred to the plurality of second fluid passageways **315** heats the steam **276** beyond a saturation temperature (superheating) such that minimal thermal energy goes towards evaporating liquid since the steam **276** has already been separated from the water **274**. With reference to FIG. **2**, the steam **276** may be discharged from the plurality of second fluid passageways **315** via the second fluid outlet **335** to the steam turbine **210**, then injected into working gas flow path **33**, such as into the combustor **26**.

[0106] FIG. **6**A is a perspective view of the heat exchanger assembly **300** of FIG. **3** according to an exemplary embodiment of the present disclosure. FIG. **6**B is a cross-sectional view of the heat exchanger assembly **300** of FIG. **6**A according to an exemplary embodiment of the present disclosure.

[0107] In at least one example embodiment, the turbine engine **10** includes the heat exchanger assembly **300** disposed in an airflow passage. For example, the heat exchanger assembly **300** may be disposed within the bypass airflow passage **56** shown in FIGS. **1**-**2**. In such embodiments, the heat exchanger assembly **300** may be positioned between the outer casing **18** and the nacelle **50** of the turbine engine **10**. Moreover, the heat exchanger assembly **300** may be incorporated into the steam system **100** of the turbine engine **10** as the boiler **202**, as described above with respect to FIG. **2**.

[0108] In at least one example embodiment, the heat exchanger assembly **300** includes a plurality of plates **600**. As shown in FIGS. **6**A-**6**B, the heat exchanger assembly **300** may include an annular heat exchanger disposed in the bypass airflow passage **56**. In such embodiments, the plurality of plates **600** may be disposed circumferentially within the bypass airflow passage **56**. Each of the plurality of plates **600** may include the first fluid channel **305**, the plurality of first fluid passageways **310**, the plurality of second fluid passageways **315**, and the separator **325**, as discussed above with respect to FIG. **3**. The first fluid channel **305** of each of the plurality of plates **600** may be adjacent the outer casing **18** and the plurality of second fluid passageways **315** of each of the plurality of plates **600** may be adjacent the nacelle **50**. Accordingly, the plurality of first fluid passageways **310** of each of the plurality of plates **600** may be between the first fluid channel **305** and the plurality of second fluid passageways **315**.

[0109] With reference to FIG. **6**B, the heat exchanger assembly **300** defines a plurality of fluid passageways **605** between the plurality of plates **600**. The plurality of fluid passageways **605** may be in fluid communication with the steam system **100**. For example, the heat exchanger assembly **300** may be incorporated as the boiler **202**, as shown in FIG. **2**, and the plurality of fluid passageways **605** may be configured to receive the combustion gases **66**. The combustion gases **66** flow through the plurality of fluid passageways **605** and transfer heat to the first fluid and the second fluid flowing through the plurality of first fluid passageways **310** and the plurality of second fluid passageways **315**, as discussed above with respect to FIGS. **2-3**.

[0110] FIG. **7**A is a perspective view of a vapor generation system according to an exemplary embodiment of the present disclosure. FIG. **7**B is a cross-sectional view of the vapor generation system of FIG. **6**A according to an exemplary embodiment of the present disclosure.

[0111] A vapor generation system, such as a heat exchanger assembly **700**, may be configured in a similar manner as the exemplary heat exchanger assembly **300** discussed above with respect to FIGS. **6**A-**6**B. For example, the heat exchanger assembly **700** may be disposed within an airflow passage of the turbine engine **10**, such as the bypass airflow passage **56**. In such embodiments, the heat exchanger assembly **700** may be between the outer casing **18** and the nacelle **50** of the turbine engine **10**. Moreover, the heat exchanger assembly **700** may be incorporated into the steam system **100** of the turbine engine **10** as the boiler **202**, as described above with respect to FIG. **2**.

[0112] In at least one example embodiment, the heat exchanger assembly **700** includes a plurality of plates **705**. As shown in FIGS. **7**A-**7**B, the plurality of plates **705** may be arranged in a stacked relationship. For example, the plurality of plates **705** may be stacked between a first side **711** and a second side **712** opposite the first side **711** of the heat exchanger assembly **700**. The first side **711** may be adjacent the outer casing **18** and the second side **712** may be adjacent the nacelle **50**.

[0113] In at least one example embodiment, the plurality of plates **705** include the first fluid channel **305**, the plurality of first fluid passageways **310**, and the plurality of second fluid passageways **315**. More specifically, the first fluid channel **305** may include at least one of the plurality of plates **705** positioned adjacent the first side **711**, the plurality of second fluid passageways **315** may include one or more of the plurality of plates **705** adjacent the second side **712**, and the plurality of first fluid passageways **310** may include one or more of the plurality of plates **705** between the plurality of plates **705** including the first fluid channel **305** and the plurality of second fluid passageways **315**. Although FIGS. **7**A-**7**B illustrate one of the plurality of plates **705** for the first fluid channel **305** and two of the plurality of plates **705** for each of the plurality of first fluid passageways **310** and the plurality of second fluid passageways **315**, it should be understood that the heat exchanger assembly **700** may include any number of the plurality of plates **705** for each of the first fluid channel **305**, the plurality of first fluid passageways **310**, and the plurality of second fluid passageways **315**.

[0114] With reference to FIG. **7**B, the heat exchanger assembly **700** defines a plurality of fluid passageways **710** between the plurality of plates **705**. The plurality of fluid passageways **710** may be in fluid communication with the steam system **100**. For example, the heat exchanger assembly **700** may be incorporated as the boiler **202**, as shown in FIG. **2**, and the plurality of fluid passageways **710** may be configured to receive the combustion gases **66**. The combustion gases **66** flow through the plurality of fluid passageways **710** and transfer heat to the first fluid and the second fluid flowing through the plurality of first fluid passageways **310** and the plurality of second fluid passageways **315**, as discussed above with respect to FIGS. **2-3**.

[0115] In at least one example embodiment, the heat exchanger assembly **700** may include a plurality of heat exchanger assemblies. In such embodiments, the plurality of the heat exchanger assemblies **700** may be disposed circumferentially within and about the bypass airflow passage **56**.

[0116] FIG. **8**A is a perspective view of another vapor generation system of the turbine engine **10** of FIGS. **1-2** according to an exemplary embodiment of the present disclosure. FIG. **8**B is a cross-sectional view of the vapor generation system of FIG. **8**A according to an exemplary embodiment of the present disclosure. FIG. **8**C is a cross-sectional view of the vapor generation system of FIG. **8**A according to an exemplary embodiment of the present disclosure.

[0117] In at least one example embodiment, the vapor generation system of the steam system **100** includes a heat exchanger assembly **800**. The heat exchanger assembly **800** may be configured in a similar manner as the exemplary heat exchanger assembly **300** discussed above with respect to FIGS. **3-5**B. For example, the heat exchanger assembly **800** includes, at least, the plurality of separation passageways **400** and the plurality of second fluid passageways **315**, which may be similar or analogous to the plurality of separation passageways **400** and the plurality of second fluid passageways **315** discussed above with respect to FIGS. **3-5**B. Moreover, the heat exchanger assembly **800** may be used in place of the heat exchanger assembly **300** in some example embodiments.

[0118] In at least one example embodiment, the heat exchanger assembly **800** includes the plurality of separation passageways **400** extending between the first separator side **401** and the second separator side **402**. The plurality of separation passageways **400** may be similar or analogous to the plurality of separation passageways discussed above with respect to FIGS. **4**A-**5**B. With reference to FIGS. **8**B-**8**C, each of the plurality of separation passageways **400** include the swirler **518** disposed adjacent the second separator side **402** and the tapered portion **505** adjacent the first separator side **401**. The swirler **518** includes the plurality of vanes **510** extending circumferentially about the separator hub **515**. The swirler **518** is configured to add swirl to a vapor-liquid mixture, such as the first fluid and the second fluid, received from the plurality of first fluid passageways **310** such that the first fluid and the second fluid may be at least partially separated. For example, the first fluid, such as the water **274**, in the form of liquid droplets may flow radially outward towards the interior walls of the plurality of separation passageways **400**. The first fluid in the form of liquid droplets or liquid film formed from the swirl may be discharged through the plurality of fluid openings **408** defined by the tapered portion **505** of the plurality of separation passageways **400** to a fluid cavity **805**. For example, each of the plurality of fluid openings **408** are in fluid communication with the fluid cavity **805**.

[0119] In at least one example embodiment, the fluid cavity **805** is adjacent the first separator side **401** and defined between the tapered portion **505** of each of the plurality of separation passageways **400**. The fluid cavity **805** is in fluid communication with a re-boiling cavity **810** adjacent the first separator side **401**. The re-boiling cavity **810** may form a leading edge of the heat exchanger assembly **800** such that the fluid cavity **805** is between the re-boiling cavity **810** and the plurality of separation passageways **400**. Moreover, a plurality of fluid crossover pathways **815** may be defined between the fluid cavity **805** and the re-boiling cavity **810**. The first fluid, such as the liquid droplets, may flow from the fluid cavity **805** to the re-boiling cavity **810** via the plurality of fluid crossover pathways **815**. For example, the plurality of fluid crossover pathways **815** may include a wicking material for directing the first fluid from the fluid cavity **805** to the re-boiling cavity **810**.

[0120] In at least one example embodiment, a number of the plurality of fluid crossover pathways **815** is the same as a number of the plurality of separation passageways **400**. For example, at least one of the plurality of fluid crossover pathways **815** may be associated with each of the plurality of separation passageways **400**. In other example embodiments, the plurality of fluid crossover pathways **815** may be between the tapered portion **505** of each of the plurality of separation passageways **400**. For example, one of the plurality of fluid crossover pathways **815** may be between the tapered portion **505** of two adjacent ones of the plurality of separation passageways **400** such that the number of the plurality of fluid crossover pathways **815** is one less than the number of the plurality of separation passageways **400**.

[0121] The second fluid is discharged from the second fluid outlet **430** of the plurality of separation passageways **400** into a vapor cavity **820**. The vapor cavity **820** may be adjacent the first separator side **401**. More specifically, the vapor cavity **820** may be between the fluid cavity **805** and the re-boiling cavity **810**. Additionally, the plurality of fluid crossover pathways **815** may be at least partially disposed in the vapor cavity **820** and fluidly isolated from the vapor cavity **820**. Moreover, the vapor cavity **820** is in fluid communication with the plurality of second fluid passageways **315** via a vapor manifold **825**. The vapor manifold **825** defines a fluid channel through which the second fluid, or vapor, may flow from the vapor cavity **820** to the plurality of second fluid passageways **315**.

[0122] In at least one example embodiment, the re-boiling cavity **810** is configured to heat the first fluid received from the fluid cavity **805** via the plurality of fluid crossover pathways **815**. For example, the combustion gases **66** are configured to flow through the heat exchanger assembly **800** in a similar manner as the heat exchanger assembly **300**, as discussed above with respect to FIG. **2**. Because the re-boiling cavity **810** is adjacent the first separator side **401**, such that it forms a leading edge of the heat exchanger assembly **800**, the combustion gases **66** come into contact with the re-boiling cavity **810** and transfer heat to the first fluid therein. Accordingly, the first fluid is heated within the re-boiling cavity **810** and the second fluid, such as the steam **276**, is generated.

[0123] With reference to FIGS. **8**B-**8**C, the re-boiling cavity **810** defines a tortuous pathway trough which the first fluid and the second fluid flow. For example, a plurality of baffles **830** may be disposed in the re-boiling cavity **810**. As shown in FIGS. **8**B-**8**C, the plurality of baffles **830** may be disposed in an alternating arrangement defining the tortuous pathway. The tortuous pathways defined between the plurality of baffles **830** is configured to increase the liquid residence time within the re-boiling cavity **810** and enhance liquid droplet coalescence.

[0124] In at least one example embodiment, the re-boiling cavity **810** generates a liquid-vapor mixture including the first fluid and the second fluid. The liquid-vapor mixture is discharged from the re-boiling cavity to a mist eliminator **835**. As shown in FIGS. **8**A-**8**C, the mist eliminator **835** may be between the re-boiling cavity **810** and the vapor cavity **820**. In at least one example embodiment, the mist eliminator **835** includes a porous pad. In some additional example embodiments, the mist eliminator **835** includes a porous structure formed of a plurality of wires.

[0125] The mist eliminator **835** is configured to collect liquid droplets. For example, the mist eliminator **835** collects and traps the first fluid from the liquid-vapor mixture to prevent the first fluid from flowing to the vapor manifold **825** while allowing the second fluid to pass through the mist eliminator **835** to the vapor manifold **825**. Additionally, a wicking material may be disposed at least partially within the re-boiling cavity **810** for directing the first fluid from the mist eliminator **835** to the re-boiling cavity **810**.

[0126] The second fluid discharged from the re-boiling cavity **810** and the mist eliminator **835** and is combined with the second fluid discharged from the vapor cavity **820** within the vapor manifold **825**. From the vapor manifold **825**, the second fluid may flow into the plurality of second fluid passageways **315** such that the second fluid is further heated to a degree of superheat, as described above. Moreover, with reference to FIG. **2**, the second fluid (the steam **276**) may be discharged from the plurality of second fluid passageways **315** via the second fluid outlet **335** to the steam turbine **210**, then injected into working gas flow path **33**, such as into the combustor **26**.

[0127] Further aspects are provided by the subject matter of the following clauses:

[0128] A vapor generation apparatus, comprising: a fluid channel extending between a first end and a second end of the vapor generation apparatus; a plurality of first fluid passageways extending between the first end and the second end; a plurality of second fluid passageways extending between the first end and the second end, wherein the plurality of first fluid passageways are disposed between the fluid channel and the plurality of second fluid passageways; a fluid chamber adjacent the second end and configured to receive a first fluid, wherein the fluid chamber is in fluid

communication with the fluid channel and the plurality of first fluid passageways; and a separator adjacent the first end and in fluid communication with the fluid channel, the plurality of first fluid passageways, and the plurality of second fluid passageways; wherein the fluid channel is configured to receive the first fluid from the separator; and wherein the plurality of second fluid passageways are configured to receive a second fluid from the separator.
[0129] The vapor generation apparatus of any preceding clause, wherein: the plurality of first fluid passageways are configured to heat the first fluid; and the separator is configured to separate the first fluid and the second fluid.
[0130] The vapor generation apparatus of any preceding clause, wherein: the plurality of first fluid passageways are configured to heat the first fluid to generate a gas-vapor mixture including the first fluid and the second fluid; and the gas-vapor mixture is greater than or equal to 90% vapor by mass.
[0131] The vapor generation apparatus of any preceding clause, wherein: the first fluid comprises a liquid; and the second fluid comprises a vapor.
[0132] The vapor generation apparatus of any preceding clause, wherein: the fluid channel comprises a first fluid channel; the separator is adjacent the first end and the plurality of first fluid passageways; and the separator comprises: a plurality of separation passageways extending between a first separator end and a second separator end, the first separator end adjacent the first end and the second separator end adjacent the plurality of first fluid passageways, and a separation chamber adjacent the first separator end and in fluid communication with the plurality of separation passageways.
[0133] The vapor generation apparatus of any preceding clause, wherein: the separator is in fluid communication with the fluid channel via the separation chamber; and the separator is in fluid communication with the plurality of second fluid passageways via a second fluid channel adjacent the first end.
[0134] The vapor generation apparatus of any preceding clause, further comprising a wicking material disposed in the separation chamber, the wicking material configured to guide the first fluid from the separation chamber to the fluid channel.
[0135] The vapor generation apparatus of any preceding clause, wherein: a plurality of vanes are disposed circumferentially within the separator adjacent the second separator end; and each of the plurality of separation passageways define a plurality of openings adjacent the first separator end, the plurality of openings in fluid communication with the separation chamber.
[0136] The vapor generation apparatus of any preceding clause, wherein: the separator includes at least one first fluid pathway flowing from the second separator end to the first separator end, through the plurality of openings, and to the first fluid channel; and the separator includes at least one second fluid pathway flowing from the second separator end to the first separator end and to the plurality of second fluid passageways.
[0137] The vapor generation apparatus of any preceding clause, wherein each of the plurality of separation passageways comprise a tapered surface adjacent the second separator end, the tapered surface including the plurality of openings.
[0138] The vapor generation apparatus of any preceding clause, wherein the first fluid channel defines at least one inlet opening adjacent the first end fluidly coupling the separator and the first fluid channel.
[0139] The vapor generation apparatus of any preceding clause, further comprising a wicking material disposed in the first fluid channel, the wicking material configured to guide the first fluid within the first fluid channel to the fluid chamber.
[0140] The vapor generation apparatus of any preceding clause, wherein the first fluid channel, the plurality of first fluid passageways, and the plurality of second fluid passageways are arranged in a stacked relationship.
[0141] The vapor generation apparatus of any preceding clause, further comprising an inlet nozzle

fluidly coupled to the fluid chamber, the inlet nozzle configured to deliver the first fluid to the fluid chamber.
[0142] The vapor generation apparatus of any preceding clause, wherein the separator comprises: a plurality of separation passageways extending between a first separator end and a second separator end, the first separator end adjacent the first end and the second separator end adjacent the plurality of first fluid passageways; a fluid cavity in fluid communication with the plurality of separation passageways and configured to receive the first fluid; a re-boiling cavity adjacent the first separator end; a vapor cavity in fluid communication with the plurality of separation passageways and configured to receive the second fluid, the vapor cavity between the fluid cavity and the re-boiling cavity; and a plurality of fluid crossover channels disposed in the vapor cavity and configured to fluidly couple the plurality of separation passageways and the re-boiling cavity.
[0143] The vapor generation apparatus of any preceding clause, wherein: the separator includes at least one first fluid pathways flowing from the plurality of separation passageways, the fluid cavity, the plurality of fluid crossover channels, and the re-boiling cavity; and the separator includes at least one second fluid pathways flowing from the plurality of separation passageways to the vapor cavity.
[0144] The vapor generation apparatus of any preceding clause, wherein the separator further comprises a vapor manifold in fluid communication with the re-boiling cavity and the vapor cavity, the vapor manifold configured to discharge the second fluid to the plurality of second fluid passageways.
[0145] The vapor generation apparatus of any preceding clause, wherein the separator further comprises a mist eliminator in fluid communication with the re-boiling cavity and the vapor manifold, the mist eliminator configured to prevent the first fluid from flowing to the vapor manifold.
[0146] The vapor generation apparatus of any preceding clause, wherein the separator further comprises a wicking material disposed in the re-boiling cavity, the wicking material configured to guide the first fluid from the mist eliminator to the re-boiling cavity.
[0147] A turbine engine, comprising: a turbomachine comprising a compressor section, a combustion section, and a turbine section in serial flow order and together defining a working gas flow path; a steam system operable with the turbomachine; and a heat exchanger in fluid communication with the steam system, wherein the heat exchanger comprises: a fluid channel extending between a first end and a second end of the heat exchanger, a plurality of first fluid passageways extending between the first end and the second end, a plurality of second fluid passageways extending between the first end and the second end, wherein the plurality of first fluid passageways are between the fluid channel and the plurality of second fluid passageways, a fluid chamber adjacent the second end and configured to receive a first fluid, wherein the fluid chamber is in fluid communication with the fluid channel and the plurality of first fluid passageways, and a separator adjacent the first end and in fluid communication with the fluid channel, the plurality of first fluid passageways, and the plurality of second fluid passageways.
[0148] The turbine engine of any preceding clause, wherein: the fluid channel is configured to receive the first fluid from the separator; and the plurality of second fluid passageways are configured to receive a second fluid from the separator.
[0149] The turbine engine of any preceding clause, wherein: the plurality of first fluid passageways are configured to heat the first fluid; and the separator is configured to separate the first fluid and the second fluid.
[0150] The turbine engine of any preceding clause, wherein: the plurality of first fluid passageways are configured to heat the first fluid to generate a gas-vapor mixture including the first fluid and the second fluid; and the gas-vapor mixture is greater than or equal to 90% vapor by mass.
[0151] The turbine engine of any preceding clause, wherein: the first fluid comprises a liquid; and the second fluid comprises a vapor.

[0152] The turbine engine of any preceding clause, wherein: the separator is adjacent the first end and the plurality of first fluid passageways; and the separator comprises: a plurality of separation passageways extending between a first separator end and a second separator end, the first separator end adjacent the first end and the second separator end adjacent the plurality of first fluid passageways, and a separation chamber adjacent the first separator end and in fluid communication with the plurality of separation passageways.

[0153] The turbine engine of any preceding clause, wherein: the fluid channel comprises a first fluid channel; the separator is in fluid communication with the first fluid channel via the separation chamber; and the separator is in fluid communication with the plurality of second fluid passageways via a second fluid channel adjacent the first end.

[0154] The turbine engine of any preceding clause, further comprising a wicking material disposed in the separation chamber, the wicking material configured to guide the first fluid from the separation chamber to the first fluid channel.

[0155] The turbine engine of any preceding clause, wherein: a plurality of vanes are disposed circumferentially within the separator adjacent the second separator end; and each of the plurality of separation passageways define a plurality of openings adjacent the second separator end, the plurality of openings in fluid communication with the separation chamber.

[0156] The turbine engine of any preceding clause, wherein: the separator includes at least one first fluid pathway flowing from the second separator end to the first separator end, through the plurality of openings, and to the fluid channel; and the separator includes at least one second fluid pathway flowing from the second separator end to the first separator end and to the plurality of second fluid passageways.

[0157] The turbine engine of any preceding clause, wherein each of the plurality of separation passageways comprise a tapered surface adjacent the second separator end, the tapered surface including the plurality of openings.

[0158] The turbine engine of any preceding clause, wherein the first fluid channel defines at least one inlet opening adjacent the first end fluidly coupling the separator and the first fluid channel.

[0159] The turbine engine of any preceding clause, further comprising a wicking material disposed in the first fluid channel, the wicking material configured to guide the first fluid within the first fluid channel to the fluid chamber.

[0160] The turbine engine of any preceding clause, wherein the first fluid channel, the plurality of first fluid passageways, and the plurality of second fluid passageways are arranged in a stacked relationship.

[0161] The turbine engine of any preceding clause, further comprising an inlet nozzle fluidly coupled to the fluid chamber, the inlet nozzle configured to deliver the first fluid to the fluid chamber.

[0162] The turbine engine of any preceding clause, wherein the heat exchanger comprises an annular heat exchanger disposed circumferentially within the turbine engine.

[0163] The turbine engine of any preceding clause, wherein: the plurality of second fluid passageways is adjacent a nacelle of the turbine engine; and the fluid channel is opposite the plurality of second fluid passageways.

[0164] The turbine engine of any preceding clause, wherein the heat exchanger comprises a plurality of heat exchangers disposed circumferentially within the turbine engine.

[0165] The turbine engine of any preceding clause, wherein the separator comprises: a plurality of separation passageways extending between a first separator end and a second separator end, the first separator end adjacent the first end and the second separator end adjacent the plurality of first fluid passageways; a fluid cavity in fluid communication with the plurality of separation passageways and configured to receive the first fluid; a re-boiling cavity adjacent the first separator end; a vapor cavity in fluid communication with the plurality of separation passageways and configured to receive a second fluid, the vapor cavity between the fluid cavity and the re-boiling

cavity; and a plurality of fluid crossover channels disposed in the vapor cavity and configured to fluidly couple the plurality of separation passageways and the re-boiling cavity.
[0166] The turbine engine of any preceding clause, wherein: the separator includes at least one first fluid pathways flowing from the plurality of separation passageways, the fluid cavity, the plurality of fluid crossover channels, and the re-boiling cavity; and the separator includes at least one second fluid pathways flowing from the plurality of separation passageways to the vapor cavity.
[0167] The turbine engine of any preceding clause, wherein the separator further comprises a vapor manifold in fluid communication with the re-boiling cavity and the vapor cavity, the vapor manifold configured to discharge the second fluid to the plurality of second fluid passageways.
[0168] The turbine engine of any preceding clause, wherein the separator further comprises a mist eliminator in fluid communication with the re-boiling cavity and the vapor manifold, the mist eliminator configured to prevent the first fluid from flowing to the vapor manifold.
[0169] The turbine engine of any preceding clause, wherein the separator further comprises a wicking material disposed in the re-boiling cavity, the wicking material configured to guide the first fluid from the mist eliminator to the re-boiling cavity.
[0170] This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A vapor generation apparatus, comprising: a fluid channel extending between a first end and a second end of the vapor generation apparatus; a plurality of first fluid passageways extending between the first end and the second end; a plurality of second fluid passageways extending between the first end and the second end, wherein the plurality of first fluid passageways are disposed between the fluid channel and the plurality of second fluid passageways; a fluid chamber adjacent the second end and configured to receive a first fluid, wherein the fluid chamber is in fluid communication with the fluid channel and the plurality of first fluid passageways; and a separator adjacent the first end and in fluid communication with the fluid channel, the plurality of first fluid passageways, and the plurality of second fluid passageways; wherein the fluid channel is configured to receive the first fluid from the separator; and wherein the plurality of second fluid passageways are configured to receive a second fluid from the separator.
2. The vapor generation apparatus of claim 1, wherein: the plurality of first fluid passageways are configured to heat the first fluid; and the separator is configured to separate the first fluid and the second fluid.
3. The vapor generation apparatus of claim 2, wherein: the plurality of first fluid passageways are configured to heat the first fluid to generate a gas-vapor mixture including the first fluid and the second fluid; and the gas-vapor mixture is greater than or equal to 90% vapor by mass.
4. The vapor generation apparatus of claim 1, wherein: the separator is adjacent the first end and the plurality of first fluid passageways; and the separator comprises: a plurality of separation passageways extending between a first separator end and a second separator end, the first separator end adjacent the first end and the second separator end adjacent the plurality of first fluid passageways, and a separation chamber adjacent the first separator end and in fluid communication with the plurality of separation passageways.
5. The vapor generation apparatus of claim 4, wherein: the fluid channel comprises a first fluid

channel; the separator is in fluid communication with the first fluid channel via the separation chamber; and the separator is in fluid communication with the plurality of second fluid passageways via a second fluid channel adjacent the first end.

6. The vapor generation apparatus of claim 4, wherein: a plurality of vanes are disposed circumferentially within the separator adjacent the second separator end; and each of the plurality of separation passageways define a plurality of openings adjacent the first separator end, the plurality of openings in fluid communication with the separation chamber.

7. The vapor generation apparatus of claim 6, wherein: the separator includes at least one first fluid pathway flowing from the second separator end to the first separator end, through the plurality of openings, and to the fluid channel; and the separator includes at least one second fluid pathway flowing from the second separator end to the first separator end and to the plurality of second fluid passageways.

8. The vapor generation apparatus of claim 1, wherein the fluid channel, the plurality of first fluid passageways, and the plurality of second fluid passageways are arranged in a stacked relationship.

9. The vapor generation apparatus of claim 1, further comprising an inlet nozzle fluidly coupled to the fluid chamber, the inlet nozzle configured to deliver the first fluid to the fluid chamber.

10. A turbine engine, comprising: a turbomachine comprising a compressor section, a combustion section, and a turbine section in serial flow order and together defining a working gas flow path; a steam system operable with the turbomachine; and a heat exchanger in fluid communication with the steam system, wherein the heat exchanger comprises: a fluid channel extending between a first end and a second end of the heat exchanger, a plurality of first fluid passageways extending between the first end and the second end, a plurality of second fluid passageways extending between the first end and the second end, wherein the plurality of first fluid passageways are between the fluid channel and the plurality of second fluid passageways, a fluid chamber adjacent the second end and configured to receive a first fluid, wherein the fluid chamber is in fluid communication with the fluid channel and the plurality of first fluid passageways, and a separator adjacent the first end and in fluid communication with the fluid channel, the plurality of first fluid passageways, and the plurality of second fluid passageways.

11. The turbine engine of claim 10, wherein: the fluid channel is configured to receive the first fluid from the separator; and the plurality of second fluid passageways are configured to receive a second fluid from the separator.

12. The turbine engine of claim 11, wherein: the plurality of first fluid passageways are configured to heat the first fluid; and the separator is configured to separate the first fluid and the second fluid.

13. The turbine engine of claim 12, wherein: the plurality of first fluid passageways are configured to heat the first fluid to generate a gas-vapor mixture including the first fluid and the second fluid; and the gas-vapor mixture is greater than or equal to 90% vapor by mass.

14. The turbine engine of claim 10, wherein: the separator is adjacent the first end and the plurality of first fluid passageways; and the separator comprises: a plurality of separation passageways extending between a first separator end and a second separator end, the first separator end adjacent the first end and the second separator end adjacent the plurality of first fluid passageways, and a separation chamber adjacent the first separator end and in fluid communication with the plurality of separation passageways.

15. The turbine engine of claim 14, wherein: the fluid channel comprises a first fluid channel; the separator is in fluid communication with the first fluid channel via the separation chamber; and the separator is in fluid communication with the plurality of second fluid passageways via a second fluid channel adjacent the first end.

16. The turbine engine of claim 14, wherein: a plurality of vanes are disposed circumferentially within the separator adjacent the second separator end; and each of the plurality of separation passageways define a plurality of openings adjacent the second separator end, the plurality of openings in fluid communication with the separation chamber.

17. The turbine engine of claim 16, wherein: the separator includes at least one first fluid pathway flowing from the second separator end to the first separator end, through the plurality of openings, and to the fluid channel; and the separator includes at least one second fluid pathway flowing from the second separator end to the first separator end and to the plurality of second fluid passageways.
18. The turbine engine of claim 10, wherein the heat exchanger comprises an annular heat exchanger disposed circumferentially within the turbine engine.
19. The turbine engine of claim 18, wherein: the plurality of second fluid passageways is adjacent a nacelle of the turbine engine; and the fluid channel is opposite the plurality of second fluid passageways.
20. The turbine engine of claim 10, wherein the heat exchanger comprises a plurality of heat exchangers disposed circumferentially within the turbine engine.